Not All Bad Demonstrations Are Equally Bad:
Quantifying How Demonstration Failure Modes Degrade Closed-Loop Policy Performance

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https://github.com/imjbassi/demo-quality-robustness

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Abstract

Imitation learning research overwhelmingly treats model architecture as the independent variable, while demonstration quality is handled as a preprocessing concern. We invert this: we hold architecture, hyperparameters, dataset size and evaluation conditions fixed, and vary only *which kind of bad demonstration* contaminates the training set, and at what rate. Five failure modes drawn from a practical data-collection taxonomy (occlusion, accidental success, corrective flailing, truncation, and inconsistent strategy) are injected into behaviour-cloning datasets for a 2D block-pushing task at contamination rates from 0 to 100%, across three seeds, for a total of 78 controlled training runs. We find that failure modes are not interchangeable: at full contamination, corrective flailing costs roughly one percentage point of closed-loop success while accidental success costs 96. Accidental success degrades as a cliff rather than a slope, remaining indistinguishable from clean data at 75% contamination before collapsing to 0.012 at 100%. Open-loop action-prediction error appears to track closed-loop success well when pooled ($r = -0.91$) but its per-mode correlation ranges from -0.15 to -0.94, and runs with statistically indistinguishable open-loop scores span closed-loop success rates from 0.315 to 0.980. We also report a prediction that failed: contrary to our hypothesis, mixing two equally reliable strategies produced no harm beyond what the individual clonability of each strategy already explains. All code, raw results and analysis are released.

1 Introduction

Behaviour cloning is the workhorse of modern robot manipulation [1], and its central known weakness is compounding error: small action errors move the policy off the demonstration distribution, where its predictions degrade further [2]. The standard responses to this are architectural or algorithmic—better policy classes [5], or interactive data collection.

Practitioners running data-collection pipelines know something the literature under-quantifies: the demonstrations themselves fail in structured, recognisable ways. A camera is occluded mid-episode. An operator fumbles and the object lands on the goal anyway. Two operators solve the same task with incompatible strategies. An episode is cut short. These are not Gaussian noise on a clean signal; they are distinct defects with distinct structure, and quality-control effort spent on one is effort not spent on another.

Prior work has established that demonstration quality matters in aggregate [3, 4]. What is less established is a *ranking*: given finite QC budget, which defect should a pipeline prioritise? And—because closed-loop evaluation is expensive and open-loop action-prediction error is cheap—does the cheap metric even detect the damage?

This paper answers both questions in a deliberately small, fully controlled setting. Our contributions are:

1. A controlled experimental design treating demonstration quality as the independent variable, with contamination rate as a single comparable axis across five distinct failure modes.
2. A ranking of failure modes by closed-loop damage, which is not the ranking a practitioner's intuition suggests.
3. Evidence that open-loop evaluation is unreliable in a specific, characterisable way: it is well correlated with closed-loop success in aggregate and near-useless within individual failure modes.
4. A negative result on strategy inconsistency, with the control that isolates it.

2 Method

2.1 Environment

Push2D is a 2D block-pushing task implemented in vectorised NumPy. A point gripper must move a block into a goal region. Pushing was chosen over reaching deliberately: it imposes a *staging* requirement, since the gripper must position itself behind the block relative to the goal before pushing is productive. This makes compounding error consequential. A policy that drifts during approach ends up on the wrong side and pushes the block *away* from the goal, which is unrecoverable within the episode budget. This is precisely the class of failure that open-loop metrics are structurally poor at surfacing, since each individual action error is small.

Observations are 10-dimensional (gripper, block and goal positions plus two relative vectors); actions are 2D displacements clipped to $[-1, 1]$. Episodes run for at most 110 steps and terminate on success, defined as the block centre falling within 0.05 of the goal.

2.2 Scripted experts

Using scripted rather than human demonstrations is what makes the experiment controlled: corruption can be applied with exact, programmatic ground truth rather than human judgement about what counts as a bad demo.

The **primary expert** is a hand-coded orbit-then-push controller that circles the block at a safe radius until aligned, then pushes toward the goal. Measured success rate: $\mathbf{100.0\%}$.

The **alternate expert** always circles in the same rotational direction from a wider staging radius, producing a visibly different route through state space. It was explicitly tuned until it *also* reaches $\mathbf{100.0\%}$, so that mixing the two tests strategy *inconsistency* rather than strategy *quality*. This tuning turns out to matter substantially (Section 4).

The **naive controller** charges blindly at the block and never reasons about the goal; the block travels along whatever ray the gripper started on. It succeeds on approximately 6% of episodes, purely by geometric luck.

2.3 Failure modes

Each mode produces demonstrations that a real pipeline could plausibly log:

Occlusion. A contiguous window covering 35% of timesteps has its observations frozen at the last visible frame. Actions remain expert-quality—the demonstrator can see, the camera cannot. The resulting transitions have corrupted inputs and pristine labels.

Accidental success. Rollouts from the naive controller, keeping only those that reached the goal. Correct outcome, no correct behaviour anywhere in the episode.

Corrective flailing. Heavy uniform jitter on the first 12 timesteps, after which the demonstrator settles into correct motion and still succeeds.

Truncation. A clean episode cut at 70% of its length and logged anyway—what a QC filter missing a premature stop looks like.

Inconsistent strategy. Complete episodes from the alternate expert. Every one is a genuine success; the only defect is disagreement with the primary strategy about what to do in a given state.

2.4 Two design decisions

Action-level corruptions are re-simulated, not pasted. Overwriting actions in a logged trajectory without re-running the dynamics yields observations that no longer follow from the actions—a physically impossible episode. That measures label noise, not bad demonstrations. Every mode that touches actions replays the episode through the environment.

Corrupted episodes must pass the same success-based QC filter a real pipeline applies. Modes are rejection-sampled to successful episodes only. Truncation is the deliberate exception, since it *is* the filter failing.

Failure mode	ρ=0.5	ρ=1.0	Pearson r
Corrective flailing	0.972	0.963	-0.35
Truncated episodes	0.930	0.987	-0.15
Occlusion (stale obs)	0.910	0.867	-0.58
Inconsistent strategy	0.933	0.303	-0.90
Accidental success	0.965	**0.012**	-0.94
Clean baseline	*0.972*		—

Table 1: Closed-loop success rate by failure mode and contamination rate, mean over three seeds. Pearson r is the within-mode correlation between open-loop MSE and closed-loop success across all runs of that mode.

2.5 Sweep axis, policy and evaluation

The sweep axis is **contamination rate** ρ, not per-episode severity. Every dataset contains the same number of episodes; ρ of them are defective and $(1-\rho)$ are clean, with per-episode severity held fixed. This places all five modes on one comparable axis and answers the question a pipeline actually asks: what fraction of the dataset can be this kind of bad before the policy breaks?

We sweep $\rho \in \{0, 0.25, 0.5, 0.75, 0.9, 1.0\}$ across three seeds, giving 78 training runs. Every run uses an identical 2×256 ReLU MLP trained with MSE behaviour cloning for 50 epochs on 400 episodes ($\approx$13k transitions). The model is deliberately small: it is not the independent variable.

Two metrics are recorded per policy. **Open-loop MSE** is action-prediction error against the expert on held-out *clean* expert states—cheap, offline, no environment interaction. **Closed-loop success rate** runs the policy in the environment over 200 episodes with initial conditions pinned to a single seed shared across every configuration.

3 Results

The clean baseline reaches 0.972 ± 0.010 closed-loop success.

3.1 Failure modes are not interchangeable

Table 1 gives the headline comparison. At full contamination, corrective flailing costs roughly one percentage point of success while accidental success costs 96. Both are “bad demonstrations” under any practitioner taxonomy, and their costs differ by two orders of magnitude.

Corrective flailing is essentially free, and plausibly mildly beneficial: jitter-then-recover trajectories provide DAgger-like coverage of off-distribution states [2], and success at $\rho = 1.0$ (0.963) is within noise of the clean baseline. QC effort spent filtering it is wasted.

Truncation is also nearly free in this setting (0.987 at $\rho = 1.0$), because the terminal push action remains easy to extrapolate from the approach phase. We expect this result to be the least transferable to tasks with a distinct terminal phase, such as grasping.

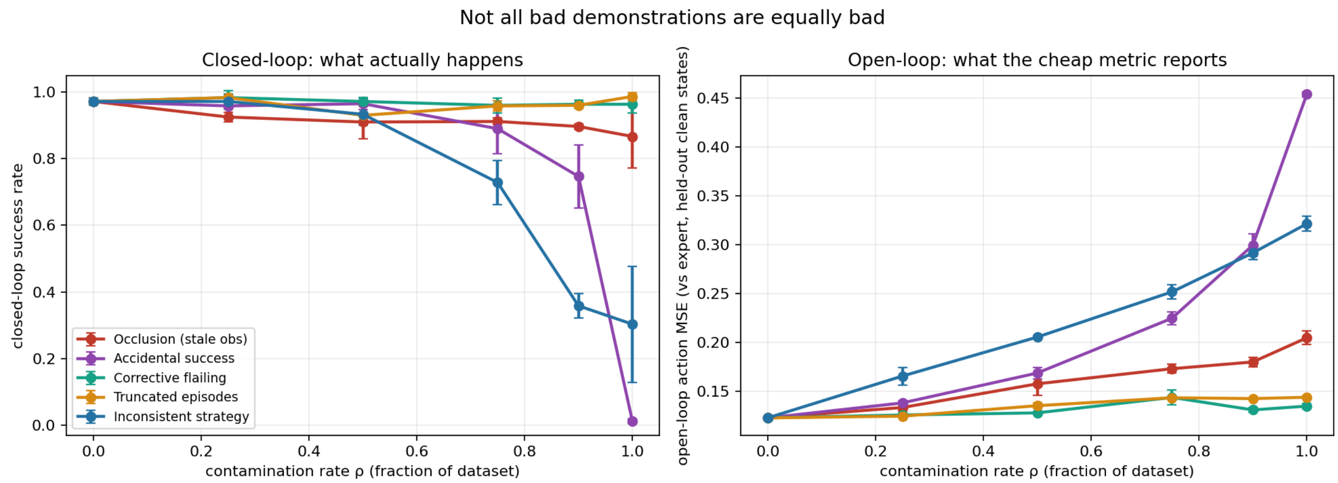

Figure 1: Dose–response curves across contamination rate. **Left:** closed-loop success rate, the quantity of interest. **Right:** open-loop MSE, the cheap proxy. Note that accidental success (purple) is indistinguishable from clean data at $\rho = 0.75$ in the left panel before collapsing at $\rho = 1.0$, and that corrective flailing and truncation (green, orange) are essentially flat in both panels. Error bars are standard deviation over three seeds.

3.2 Accidental success is a cliff, not a slope

The most operationally significant result is the shape of the accidental success curve: 0.965 at $\rho = 0.5$, 0.890 at $\rho = 0.75$, 0.747 at $\rho = 0.9$, and $\mathbf{0.012}$ at $\rho = 1.0$.

Any meaningful fraction of genuinely correct demonstrations masks the problem almost completely. A pipeline monitoring average success-labelled throughput observes nothing approaching until the policy has already failed. The diagnostic wrong-side push rate tells the same story, rising from 0.008 at $\rho = 0.5$ to 0.468 at $\rho = 1.0$: the policy has learned to shove the block in essentially arbitrary directions.

3.3 Open-loop evaluation fails in a characterisable way

Pooled across all 78 runs, open-loop MSE correlates with closed-loop success at $r = -0.91$, which would ordinarily justify using it as a screening metric.

Broken out by failure mode, the same correlation ranges from -0.15 (truncation) to -0.94 (accidental success). For the two cheapest modes the offline metric carries essentially no signal about closed-loop behaviour.

The clearest statement of the problem is a matched comparison. Among the 15 runs whose open-loop MSE falls in $[0.20, 0.30]$—indistinguishable on the offline metric—closed-loop success ranges from $\mathbf{0.315}$ to $\mathbf{0.980}$. Figure 2 shows the full picture: the pooled correlation is carried largely by a few high-MSE collapse points, while the bulk of the runs form a wide vertical band in which open-loop MSE is uninformative.

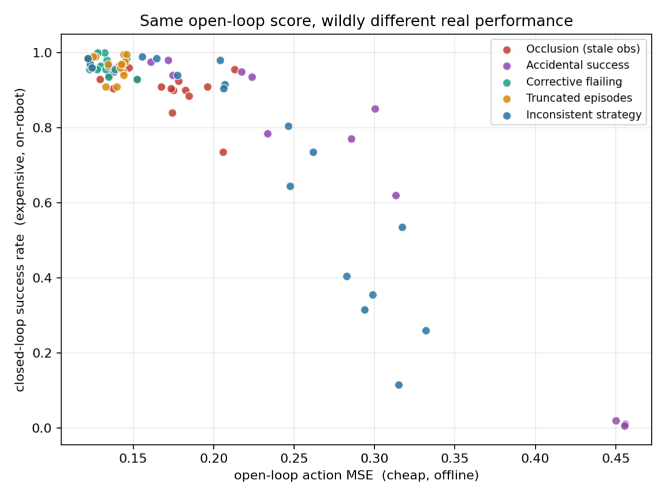

Figure 2: Open-loop MSE against closed-loop success, all 78 runs, coloured by failure mode. Points sharing an x position differ by up to 0.665 in y.

4 A prediction that failed

We predicted that mixing two valid strategies would cause multimodal-averaging collapse, since an MSE-regression policy averages conflicting action targets by construction. The data does not support this, and the control is worth stating explicitly because it nearly produced a false positive.

At $\rho = 1.0$ the dataset is entirely alternate-strategy and therefore fully self-consistent—there is no mixing left to blame. Success there (0.303) is the *clonability floor* of that strategy, not evidence of inconsistency harm. Both experts solve the task on 100% of episodes; the alternate is simply harder to clone, having a longer horizon and more orbit steps over which error compounds.

Reading intermediate ρ against the line joining the two endpoints isolates any harm attributable to mixing:

Excess harm is positive at every intermediate point. Mixing strategies was never worse than the clonability-weighted expectation; clean data protects rather than conflicts. The

ρ	Observed	Clonability-only	Excess harm
0.25	0.972	0.805	+0.167
0.50	0.933	0.637	+0.296
0.75	0.728	0.470	+0.258
0.90	0.358	0.370	−0.012

Table 2: Excess harm from mixing, relative to linear interpolation between the $\rho = 0$ and $\rho = 1$ endpoints.

inconsistent-strategy curve in Figure 1 looks damaging only because one of the two strategies is intrinsically harder to imitate.

This materially deflates our own headline number. Removing the inconsistent-strategy runs from the matched-MSE band narrows the closed-loop spread from 0.665 to approximately 0.22. That remains a real effect, and it is a considerably more modest one than the raw figure suggests.

5 Limitations

Three seeds is thin. Some cells are noisy—occlusion at $\rho = 1.0$ spans 0.735 to 0.955 across seeds. Any number quoted from this work should be replicated at ten seeds first.

A 2D toy environment. Whether these rankings survive in a 7-DOF setting with contact dynamics and visual observations is open. This work establishes a methodology, not transferable constants.

One policy class. MSE regression averages multimodal action distributions by construction. The inconsistent-strategy result in particular may be specific to this choice; replicating it with a diffusion policy [5] is the most informative single follow-up.

Transition imbalance. Alternate-strategy episodes are longer, so at fixed episode count they contribute more transitions. A transition-matched replication would tighten that arm further.

6 Conclusion

Treating demonstration quality as the independent variable, rather than an implementation detail, produces an actionable ranking: accidental success is catastrophic but only past a sharp threshold, occlusion is a steady moderate cost, and corrective flailing and truncation are close to free. The practical implication for data-collection pipelines is that QC budget is not well-allocated uniformly across defect types.

The evaluation finding is the more general one. Open-loop action-prediction error looks like a defensible screening metric in aggregate and is unreliable within failure modes—meaning that its aggregate correlation is exactly the kind of evidence that would lead a team to trust it and then be surprised. Screening a policy offline is not the same as knowing it works.

References

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